

LABORATORY

Lab 1

Radioactive Decay · United States Air Force Academy, Department of Physics and Meteorology

35 Points

NAMES

SCORE

/ 35

I · INTRODUCTION

A. Purpose

The purpose of this laboratory is to determine the half-life for a radioactive source using counting statistics, logarithm plotting, and linear regression (a.k.a. "finding the equation of a line").

You will do most of your analysis in the provided spreadsheet, but only this document will be graded.

B. Theory

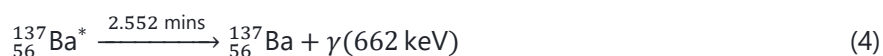
The activity of a source is the number of radioactive nuclei that decay per unit time, a value that is determined by measuring the number of particles emitted by the source per unit time. As a radioactive source decays, the activity decreases. Measuring how the activity changes versus time allows the determination of the half-life of the material. For example, the activity of a sample with a single radioactive component follows an exponential function. If the initial activity is A_0 , then the activity at any later time $A(t)$ may be expressed as:

$$A(t) = A_0 e^{-\lambda t} \quad (1)$$

where the decay constant λ is related to the half-life $t_{1/2}$ by the relationship

$$t_{1/2} = t_H = \ln(2)/\lambda \quad (2)$$

In this laboratory you will investigate the nuclear decay of an excited state of ^{137}Ba , called $^{137}\text{Ba}^*$. The $^{137}\text{Ba}^*$ is produced by the decay of ^{137}Cs , as shown in the reaction equation below.



We will separate the barium from the cesium with a simple chemical process. The sample you receive will contain only $^{137}\text{Ba}^*$, so it is the half-life of the $^{137}\text{Ba}^*$ decay that you will determine in this lab. In a Geiger-Mueller tube, any radiation which enters the tube ionizes some of the gas atoms in the tube and creates an electrical pulse which is detected by the timer/counter unit.

The statistical nature of nuclear counting experiments allows us to say that the uncertainty in the number of counts equals the square root of the counts. This only applies to the raw data.

$$\sigma_{counts} = \sqrt{N_{counts}} \quad (5)$$

II · EXPERIMENT

A. Equipment

- Geiger-Mueller detector
- Timer/counter unit
- Radioactive $^{137}\text{Ba}^*$ source, obtained from instructor

B. Procedure

1. Connect the cable from the Geiger-Mueller (GM) tube to the back of the counter unit and plug the unit into an outlet. **Do not bring anything into contact with the bottom of the GM tube, it is very fragile.**
2. Set the power supply voltage on the counter to the value on the GM tube (860 V if not labeled).
3. Obtain a radioactive $^{137}\text{Ba}^*$ source from your instructor, and place the source on the plastic tray in the top slot below the GM tube.

HINT The $^{137}\text{Ba}^*$ source has a fairly short half-life, so proceed with the counting immediately.

4. Using the built-in counter, count for 30 seconds. After recording the value, reset the counter and wait for 30 seconds, then count again. (30 second count, 30 second pause). Repeat until you have made 9 measurements. The start times for each counting interval will be 1 minute apart, and you should record each value in the table beside the *start* value for the count.

HINT Wait for exactly 30 seconds in between counts — this is very important for statistical analysis!

3 pts

TIME	COUNTS	TIME	COUNTS	TIME	COUNTS
0		3		6	
1		4		7	
2		5		8	

5. Now record a background activity measurement. Without the source present (or nearby), count the background activity for 3–5 minutes and record your data below.

1 pt

Total background counts: _____ counts

Counting time: _____ minutes

6. Using 2 points from your counting data, calculate an initial estimate for the half-life of $^{137}\text{Ba}^*$.

3 pts

7. Approximate the uncertainty in this half-life. (Use eq'n (5) to find the uncertainty in each of the 2 count values from (6), and use these to calculate the largest and smallest half-lives within the range.)

3 pts

III · ANALYSIS

A. Spreadsheet Setup

1. Open the provided spreadsheet file, read the "Instructions" tab.
2. Enter your activity vs. time data (9 data points) and your background data into the spreadsheet in Block 1 "Source Activity Data" and Block 2 "Background Activity Data."
3. Put the half-life and error you approximated into Block 3 "Fitting Parameters." What is the "Goodness of Fit" (chi-squared) value for your estimated half-life? Put this, and your solutions from (6) and (7) above, into the table.

2 pts

INITIAL ESTIMATE

Half-life (minutes)

Error (+/-) (minutes)

Goodness of Fit

4. The spreadsheet will generate a semi-log plot of activity vs. time using fitting parameters that you specify. The same data is plotted on a semi-log plot (left) and a regular plot (right).
5. Adjust the half-life and then the error value to create an estimated "best fit" to the data (make the "goodness of fit" value as close to 1 as possible, and make the error bars surround 2/3 of your data points). What are your new values?

2 pts

MANUAL FIT

Half-life (minutes)

Error (+/-) (minutes)

Goodness of Fit

B. Regression Analysis

So far, we have been estimating the half-life by minimizing error manually. We'll now use Excel's built-in tools for regression analysis. Starting in Block 4 your spreadsheet, the data is plotted again for both semi-log and standard views. Follow the steps below to find a new estimate for the half-life.

1. Use Excel's trend line to create and display a linear fit for the left chart in Block 3, and an exponential fit for the right chart. Remember to display the trend line equation and the R^2 value on the chart (options at the bottom of the trendline bar). Write the equations below.

2 pts

2. Above the semi-log plot in Block 4, input the slope, intercept, and R^2 value from the semi-log plot in Block 3. It will automatically compute a half-life. Put all of the values of your half-life and uncertainty below. How do they compare? If there are discrepancies, can you figure out why?

COMPARISON OF
RESULTS

HALF-LIFE (MINS)

ERROR (+/-) (MINS)

Initial Estimate

Manual Fit

Automatic Fit

3 pts

IV · DISCUSSION QUESTIONS

3 pts **Question 1.** The accepted half-life of Barium 137 is 2.552 minutes. Are your values consistent with this accepted value? Justify your answer by bounding your calculated half-lives with the uncertainty of your calculated half-lives.

3 pts **Question 2.** Linearize Equation (1) using the natural logarithm and show that the slope of the semi-log plot is -1 times the decay constant.

2 pts **Question 3.** What is the intercept of the previous linearized equation? (value and meaning) Confirm the value from your equations is correct on your plot.

2 pts **Question 4.** Compare the values in the semi-log best fit equation and the standard best-fit equation. How do they relate?

2 pts

Question 5. Why do we use a semi-log plot for our analysis rather than the standard plot? What are some benefits?

2 pts

Question 6. What are the 3 main methods for decreasing your exposure to radiation? Which did we use in this lab?

2 pts

Question 7. Why did we count the background radiation? How was this used in the spreadsheet?

TURN IN

Send your spreadsheet to your instructor directly, and upload your completed lab write-up to Gradescope (make sure you include all of your lab partners' names when you upload it to Gradescope — digitally link them, don't just include their names on the write-up).